Carbon price pass-through in Alberta's electricity market

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Carbon Pricing's Effect on Electricity Markets

- Research question: how do carbon pricing policy elements impact firm offer behaviour in electricity markets?
- Electricity markets, at least in jurisdictions with competitive wholesale markets, are ideal environments in which to study responses to carbon prices
- A series of policy changes in Alberta altered carbon prices, output-based credit allocations, and coverage affecting different facilities differently over time
- Rich, hourly electricity market data allows for the identification of facility- and firm-level decisions with respect to carbon price pass-through
- Extensive emissions reporting data covering almost 20 years along with hourly weather across Alberta and daily commodity prices complete the data

Results Preview

- Limited (maximum of roughly 60%) pass-through of carbon pricing costs in certain regions of the supply curve
- Physical constraints and market rules limit pass-through in lower-reaches of the hourly supply curve, leading to no pass-through in most cases
- Strategic incentives overwhelm carbon pricing pass-through in upper-reaches of the hourly supply curve, leading to limited pass-through
- Weak evidence of pass-through reaction to overall market tightness
- Caveat: offer choices are constrained at the upper and lower reaches of the merit order, and Tobit ML estimates not yet completed

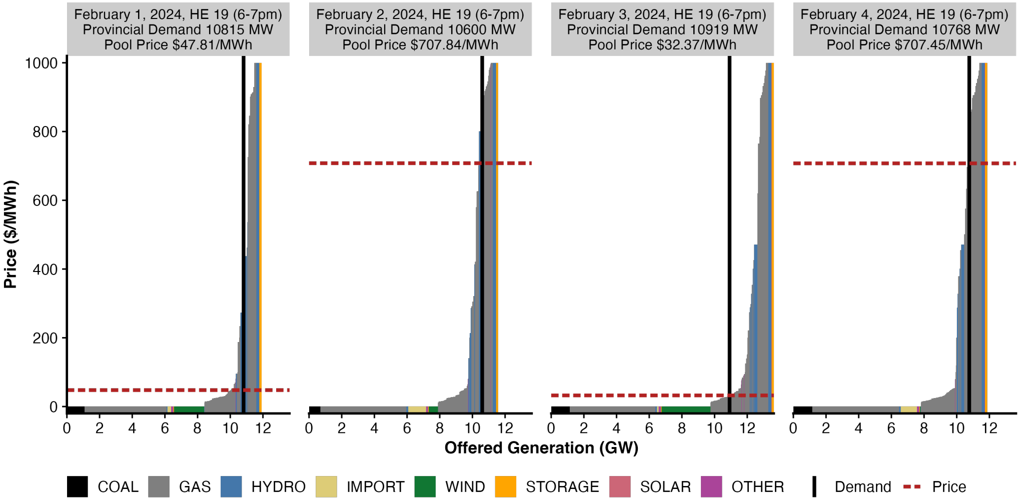
Literature on Carbon Price Pass-Through

- Fabra and Reguant (2014) finds pass-through of 110% of costs in peak-hours and roughly 60% in off-peak hours
- Fabra and Reguant (2014) finds that lump-sum credit allocations were not distortionary in the short run
- Hintermann (2016) finds pass-through rates of 81-111% depending on the hour of the day, and also finds that costs are fully passed-through on average.
- Kara et al. (2008) finds pass-through of 74% in annual electricity prices in the Nordic area.
- Nazifi (2016) shows near-complete pass-through of Australian emissions prices to wholesale power prices
- Jouvet and Solier (2013) finds insignificant pass-through in many hours in the EU.
- Brown et al. (2018) looks at the dynamics of the policies I study, but in a simplified, simulation model

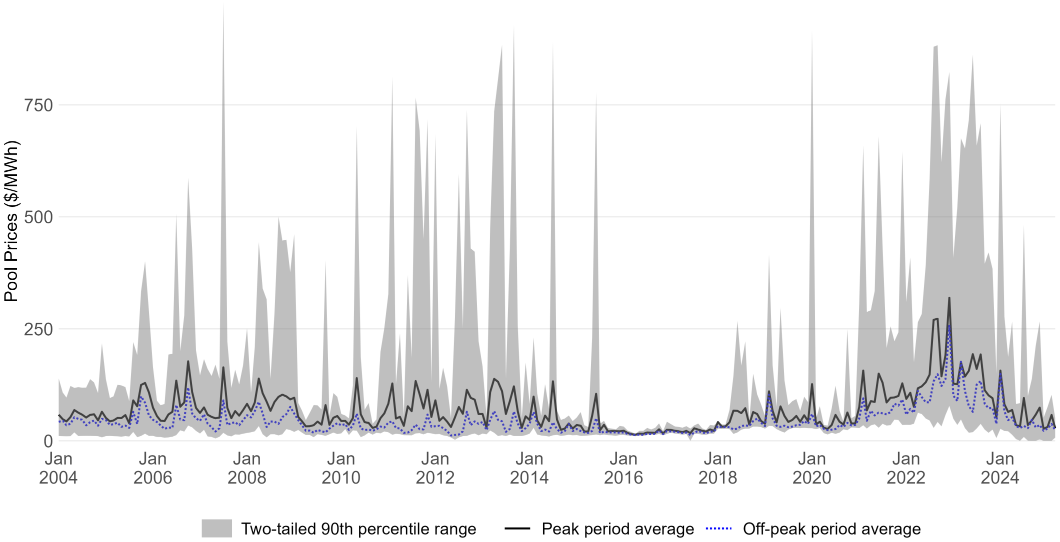
Why Alberta? Other than the obvious...

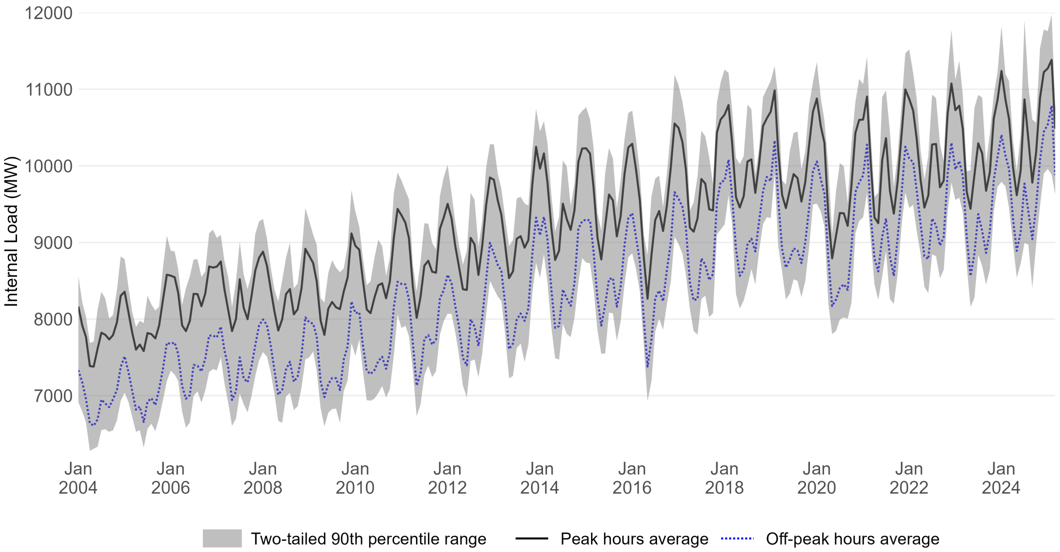
- Alberta's power market is relatively small, isolated, and features a mix of generation technologies
- Transparent wholesale market offer process, and significant hourly variability in intertie capacity and renewable generation, helps with identification
- Substantial variation in policies: three provincial regulatory regimes (2007, 2018, 2020) and one federal regime (2018) with increases in prices in many years (2016, 2017, 2020-2025)
- Policies varied in prices, coverage, and the formulae governing the issuance of output-based allocations as well
- Provincial and federal emissions reporting and compliance data are public
- A personal connection to many of the policies, in particular Alberta's *CCIR* implemented in 2018 and the federal carbon pricing regime

How does Alberta's Wholesale Power Market Work?



Source: AESO Data, graph by Andrew Leach.

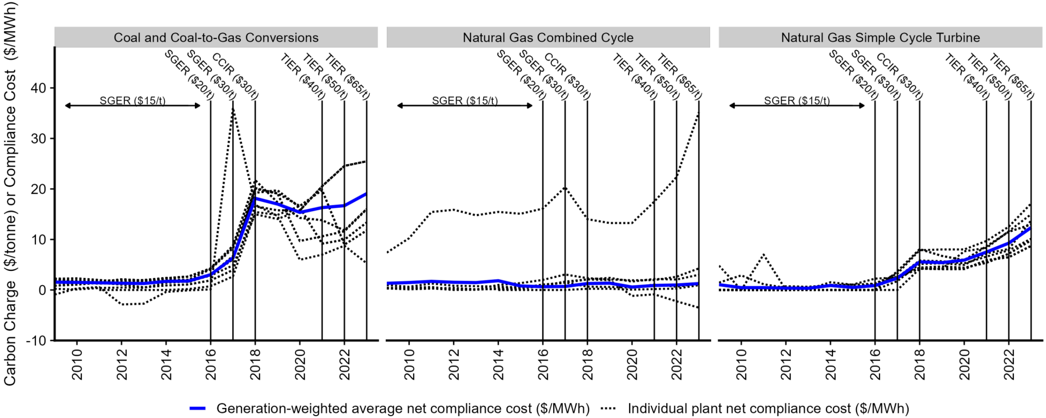

Pool Prices ($/MWh)
750
500
250
0
Jan 2004
Jan 2006
Jan 2008
Jan 2010
Jan 2012
Jan 2014
Jan 2016
Jan 2018
Jan 2020
Jan 2022
Jan 2024
Two-tailed 90th percentile range
Peak period average
Off-peak period average

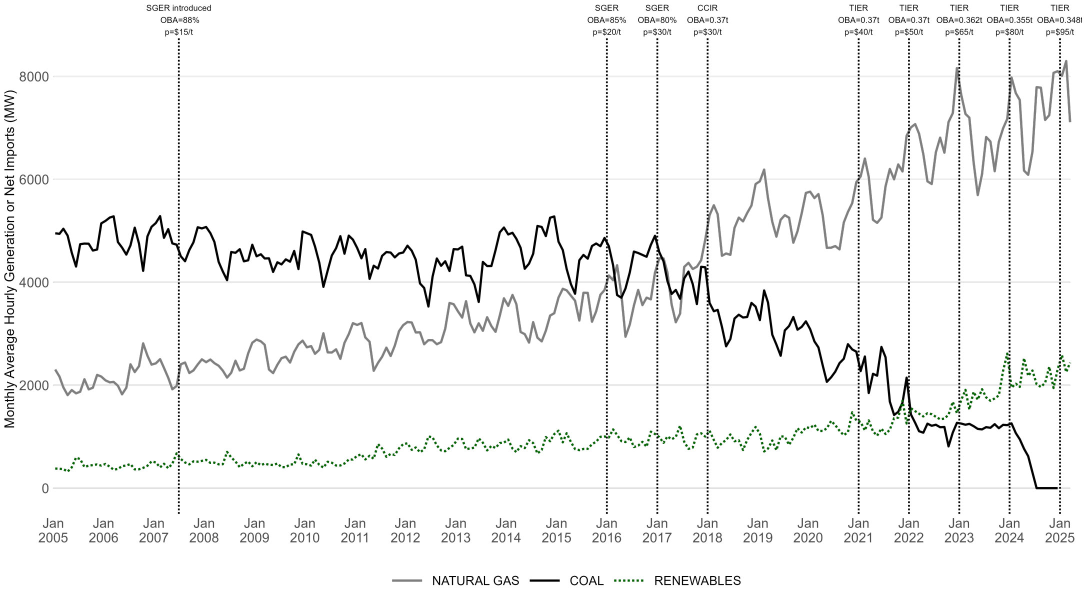

Internal Load (MW)
12000
11000
10000
9000
8000
7000
Jan 2004
Jan 2006
Jan 2008
Jan 2010
Jan 2012
Jan 2014
Jan 2016
Jan 2018
Jan 2020
Jan 2022
Jan 2024
Two-tailed 90th percentile range
Peak hours average
Off-peak hours average

Alberta GHG Emissions Policies

- *SGER* 1.0 (2007-2015): carbon price of \$15/tCO_2e, OBA at 88% of historic emissions intensity, coverage for facilities with emissions above 100,000tCO_2e/yr
- *SGER* 2.0 (2016-2017): carbon price increased to \$20/tCO_2e for 2017 and to \$30/tCO_2e for 2018, while OBA rates decline to 85% of historic emissions intensity for 2017 and 80% for 2018
- *CCIR* (2018-2019): carbon price remains at \$30/tCO_2e but OBA rates set to 0.37t/MWh for most participants. Coverage remains at 100,000 tonnes per year.
- Alberta Carbon Tax (2018-2019): carbon price set at \$30/tCO_2e expanded to cover facilities not covered by *CCIR*, with opt-in available
- *TIER* (2020-present): carbon price follows federal schedule (\$30/tCO_2e in 2020 $\rightarrow$ \$95/tCO_2e today), with coverage and OBA same as *CCIR* for electricity sector.
- Federal carbon tax (2020-2024): carbon price follows federal schedule (\$30/tCO_2e in 2020 $\rightarrow$ \$95/tCO_2e today), applies to entities not covered by *TIER*

Variation in Compliance Costs

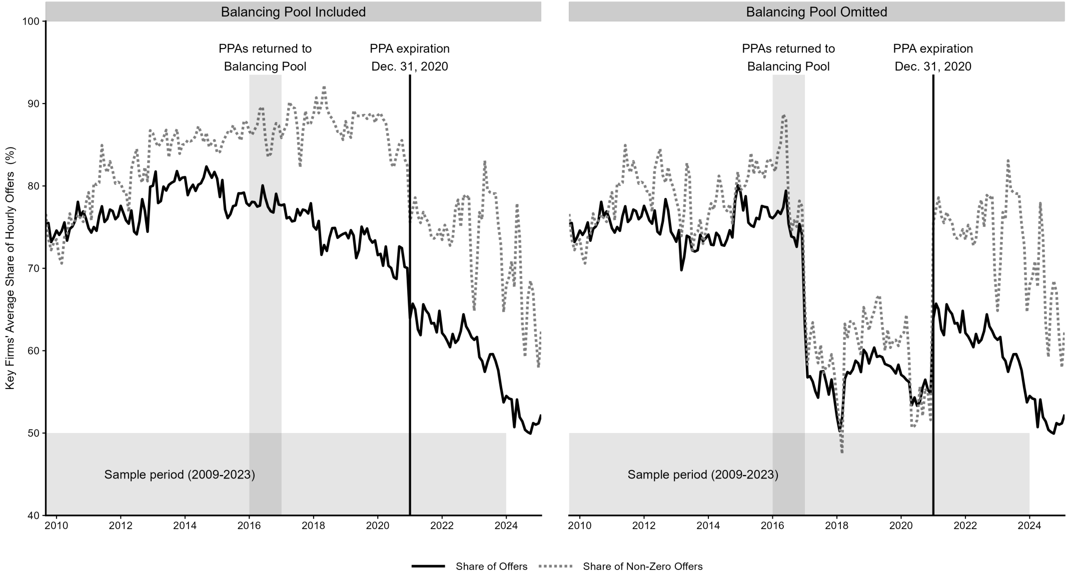




SGER introduced
OBA=88%
p=$15/t
SGER
OBA=85%
p=$20/t
SGER
OBA=80%
p=$30/t
CCIR
OBA=0.37t
p=$30/t
TIER
OBA=0.37t
p=$40/t
TIER
OBA=0.37t
p=$50/t
TIER
OBA=0.362t
p=$65/t
TIER
OBA=0.355t
p=$80/t
TIER
OBA=0.348t
p=$95/t
Monthly Average Hourly Generation or Net Imports (MW)
8000
6000
4000
2000
0
Jan 2005
Jan 2006
Jan 2007
Jan 2008
Jan 2009
Jan 2010
Jan 2011
Jan 2012
Jan 2013
Jan 2014
Jan 2015
Jan 2016
Jan 2017
Jan 2018
Jan 2019
Jan 2020
Jan 2021
Jan 2022
Jan 2023
Jan 2024
Jan 2025
NATURAL GAS
COAL
RENEWABLES

Market Power in Alberta's Electricity System

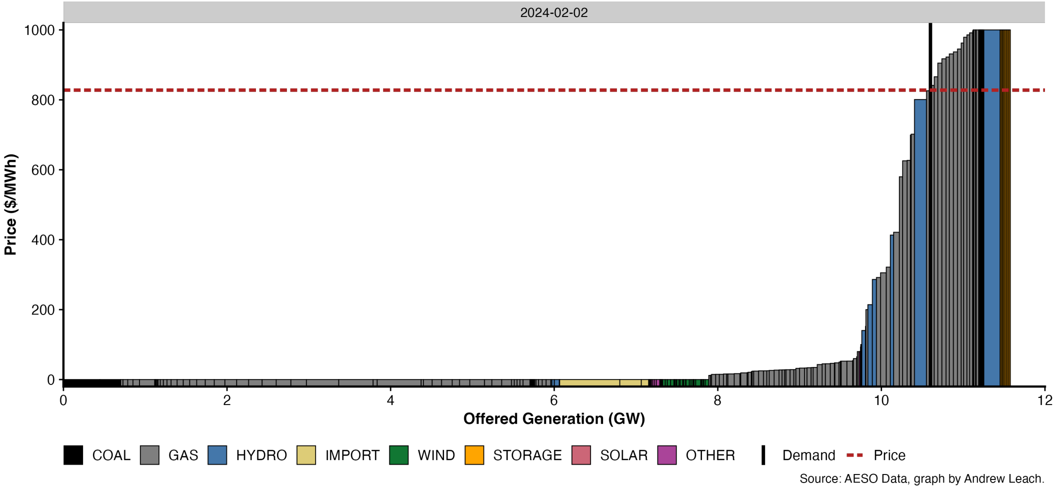
- Electricity generation was restructured in the 1990s and early 2000s
- Market was highly concentrated with legacy, regulated utility generators
- Power Purchase Arrangements (PPAs) introduced as an alternative to divestiture to mitigate market power in generation
- PPA change in law clause triggered with respect to the changes in *SGER* 2.0 and all PPAs returned to government entity (the Balancing Pool)
- Balancing Pool becomes dominant player in the market until PPAs expire at the end of 2020
- Alberta market becomes much more concentrated again, particularly among dispatchable units
- New generation construction has (had?) mitigated market power to some degree, although an acquisition in 2024-25 has the potential to return the market to a highly-concentrated state


Balancing Pool Included
Balancing Pool Omitted
PPAs returned to Balancing Pool
PPA expiration Dec. 31, 2020
Sample period (2009-2023)
Key Firms' Average Share of Hourly Offers (%)
100
90
80
70
60
50
40
2010
2012
2014
2016
2018
2020
2022
2024
Share of Offers
Share of Non-Zero Offers

Data set is comprised of 27.7 million observations linking:

- Hourly facility-level offers (up to 7 increasing-price blocks per facility, 2009-2025)
- Block-level offer control (only 2013-2025)
- Annual emissions-intensity by facility (2004-2023)
- Hourly electricity market conditions including renewable generation, trade, intertie capability, load, price, market concentration, etc. (2003-2025)
- Hourly weather in three regions of Alberta (2003-2025)
- Daily commodity price (natural gas) data (2003-2025)
- Facility characteristics (fuel, capacity, etc. 2003-2025)
- Annual emissions policy parameters including carbon prices and output-based allocations (2003-2025)

Recall the Merit Order



Estimation Strategy

- Fundamentally, what we want to do is look at individual merit order blocks and ask to what degree those blocks would have been offered at a different price under different GHG policy parameters

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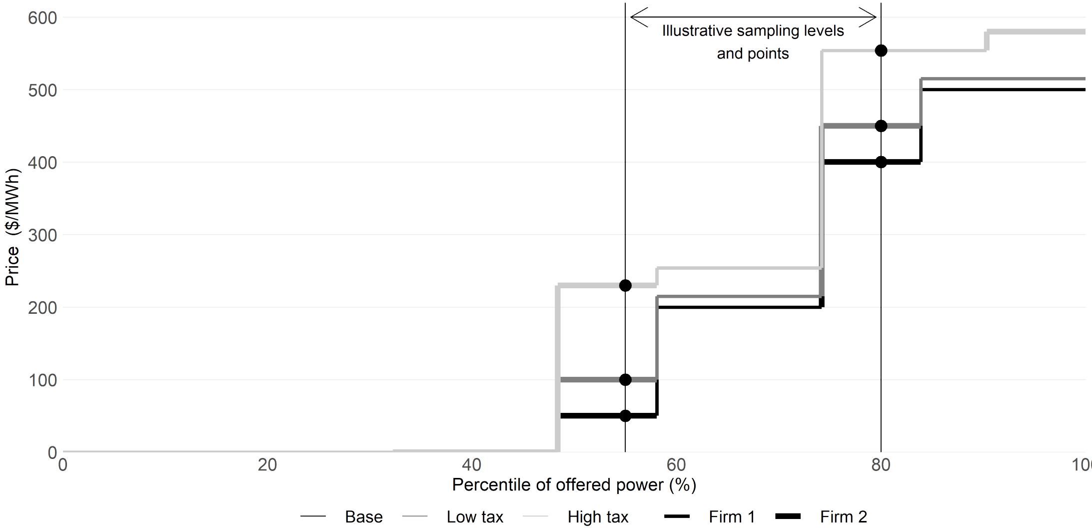
- The challenge is that the number of blocks, the size of each block, and the price at which each block is offered is a choice variable for the plant operator
- And, insofar as carbon pricing affects block size or price, position in the merit order becomes endogenous to carbon pricing


Illustrative sampling levels
and points
600
500
400
300
200
100
0
Price ($/MWh)
0
20
40
60
80
100
Percentile of offered power (%)
Base
Low tax
High tax
Firm 1
Firm 2

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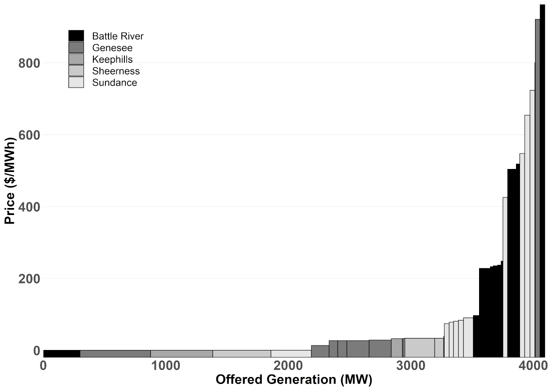
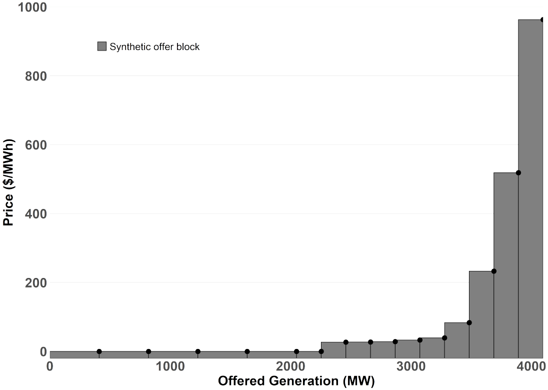
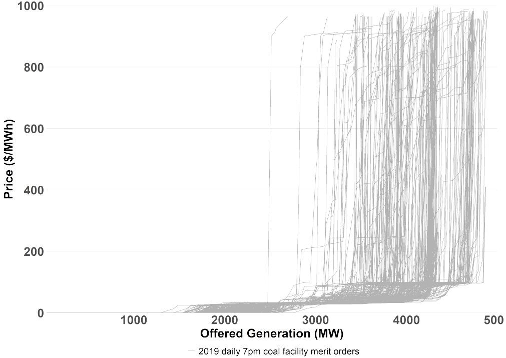
- The challenge is that the number of blocks, the size of each block, and the price at which each block is offered is a choice variable for the plant operator
- And, insofar as carbon pricing affects block size or price, position in the merit order becomes endogenous to carbon pricing
- I use an approach whereby I sample points along the supply curve at a set of percentiles of offered power in every hour in my data set
- Each sample consists of information about that point in time in the market (percentile, date, time, plant, price, offer control)
- Sampling is locally random because of the variability of wind and solar - a firm could not choose to be consistently at a given percentile in the merit order

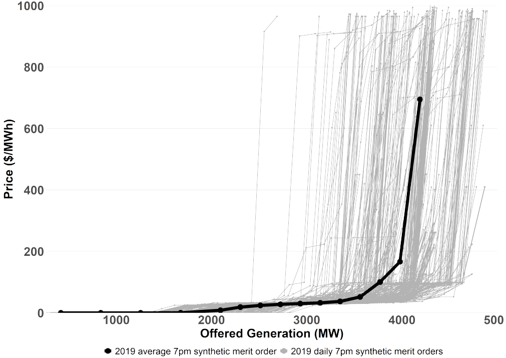
Merit Order Sampling





Merit Order Sampling with Coal Offers



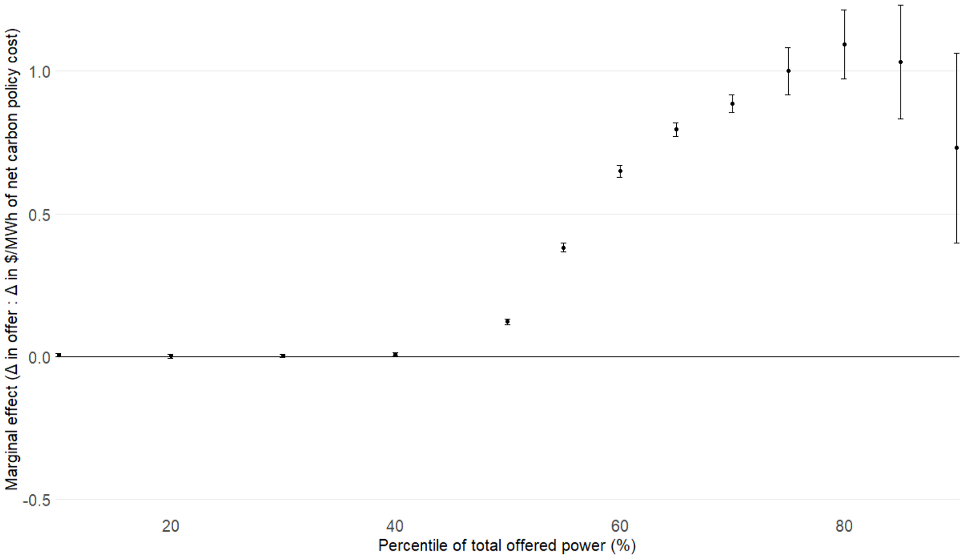


Nested Percentile Regressions

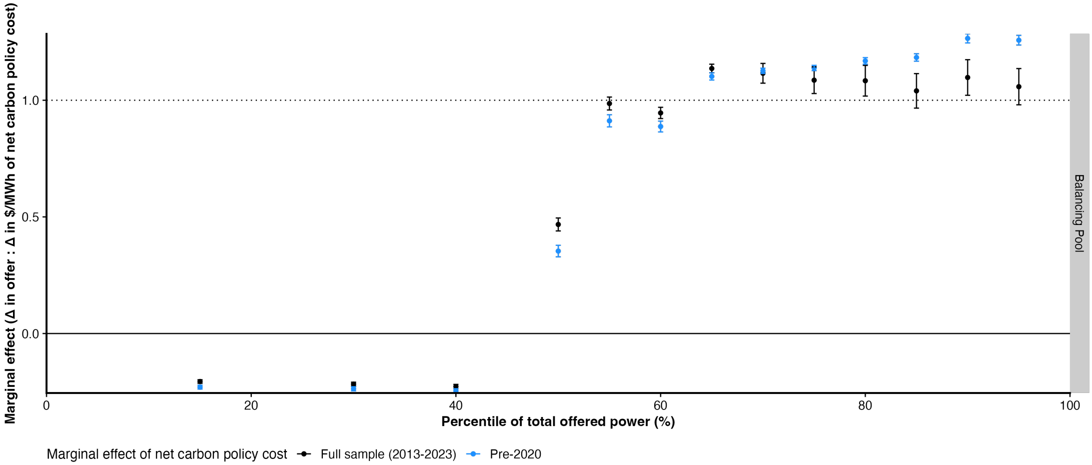
- Variation in carbon pricing over time and variation in emissions-intensity across facilities allows identification of the effect of carbon pricing on the sampled points in the merit order
- Each sampled point contains facility (F) and market (M) information, as well as carbon policy compliance costs

The estimation equation is then:

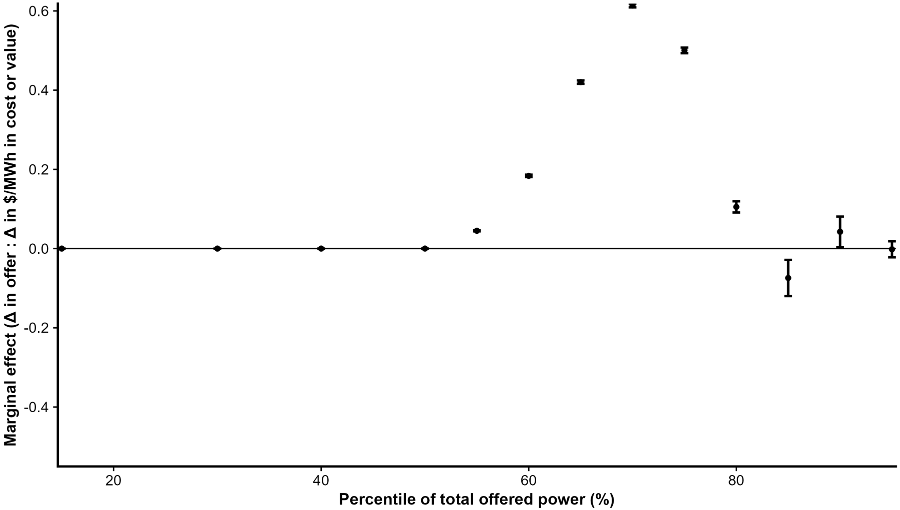
$$\underbrace{P_{j,t}}_{\substack{\text{Offer value} \\ \text{at percentile } j \\ \text{at time } t}} = \underbrace{\alpha_j}_{\substack{\text{Constant for} \\ \text{percentile } j}} + \underbrace{\beta_j M_t}_{\substack{\text{Market, policy, and} \\ \text{climate variables} \\ \text{at time } t}} + \underbrace{\delta_j F_{j,t}}_{\substack{\text{Facility} \\ \text{characteristics} \\ \text{at percentile } j \\ \text{at time } t}} + \underbrace{\zeta_j t_{j,t}}_{\substack{\text{Carbon cost} \\ \text{at percentile j} \\ \text{at time } t}} + \underbrace{\kappa_j O_{j,t}}_{\substack{\text{OBA value} \\ \text{at percentile j} \\ \text{at time } t}} + \underbrace{\epsilon_{j,t}}_{\text{Error term}}.$$


Marginal effect (Δ in offer : Δ in $/MWh of net carbon policy cost)
1.0
0.5
0.0
-0.5
20
40
60
80
Percentile of total offered power (%)

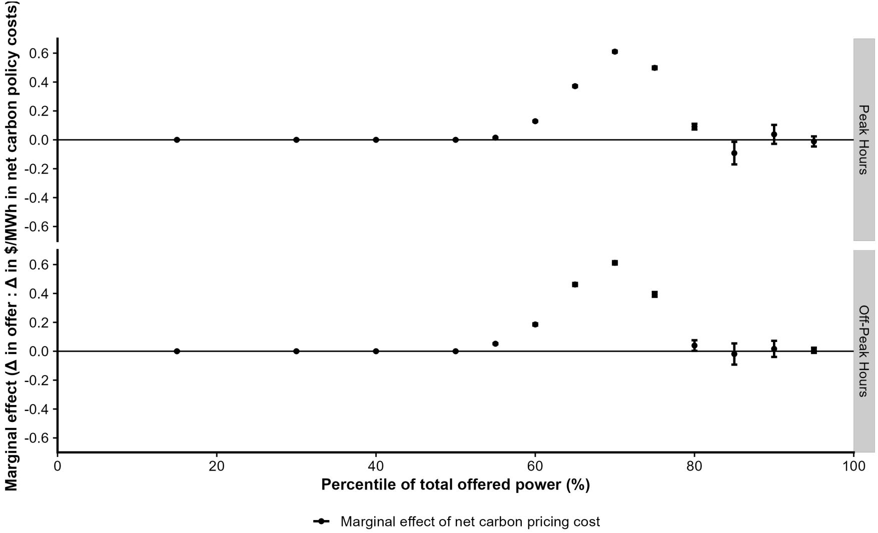
Close...in a few cases



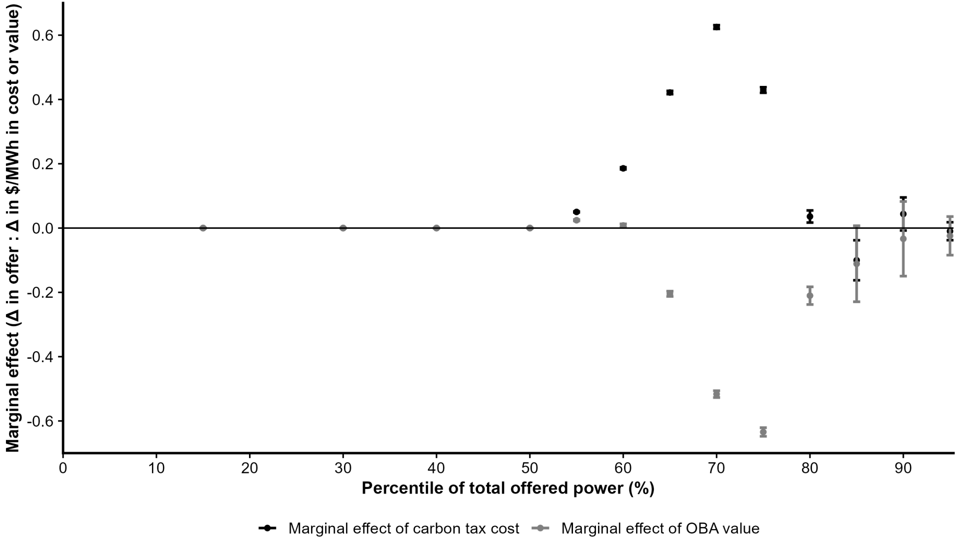
Net impact of carbon pricing, all firms, all hours



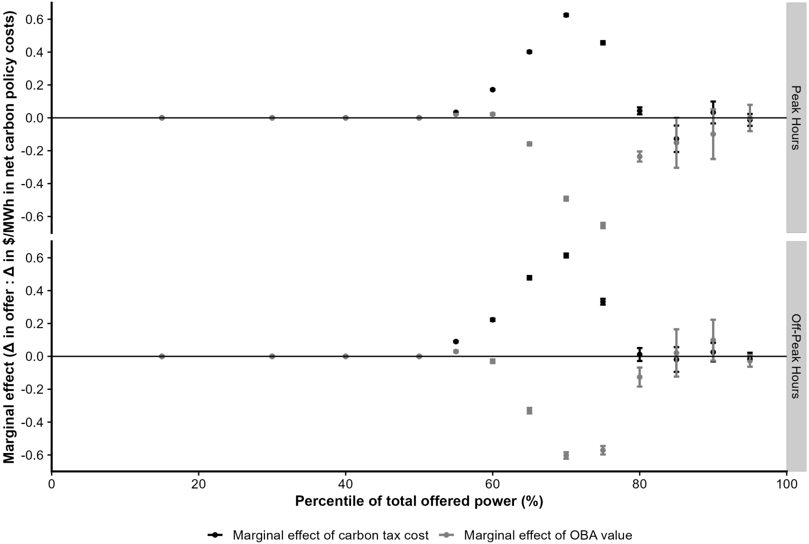
Net impact of carbon pricing, all firms, peak vs. off-peak hours



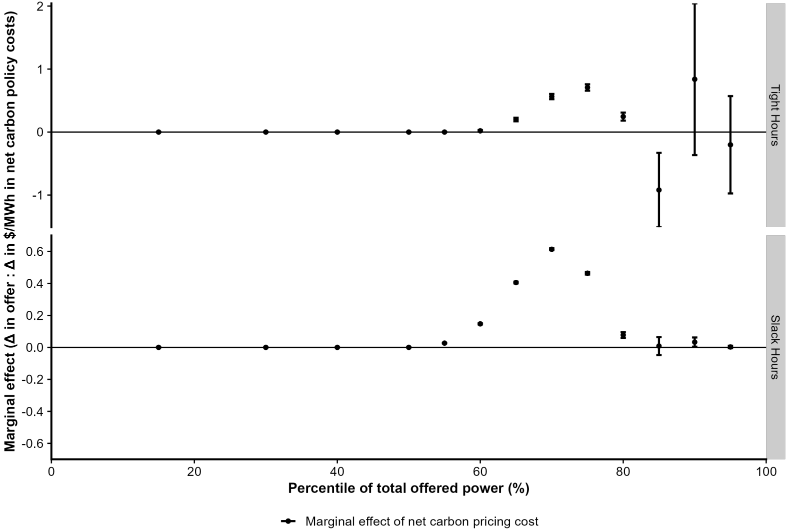
Decomposition of carbon tax and OBA, all firms, all hours



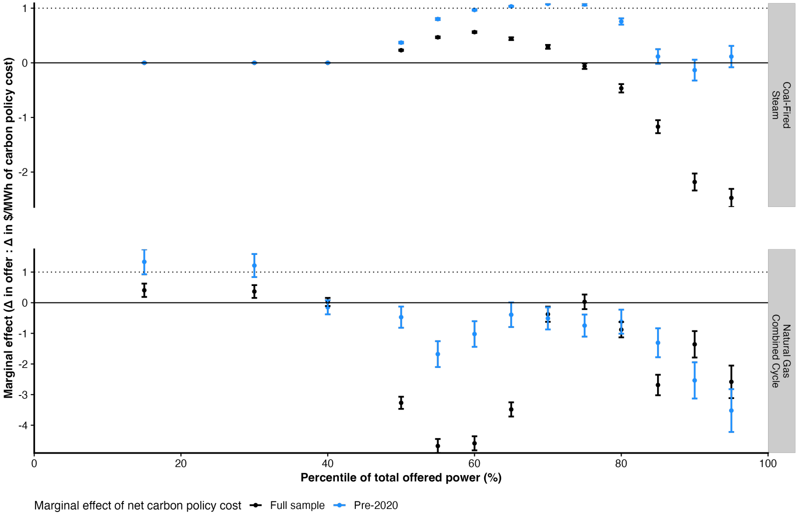
Decomposed impacts, all firms, peak vs. off-peak hours



Net impact, all firms, tight vs. slack hours



Coal and Gas Plant Offers

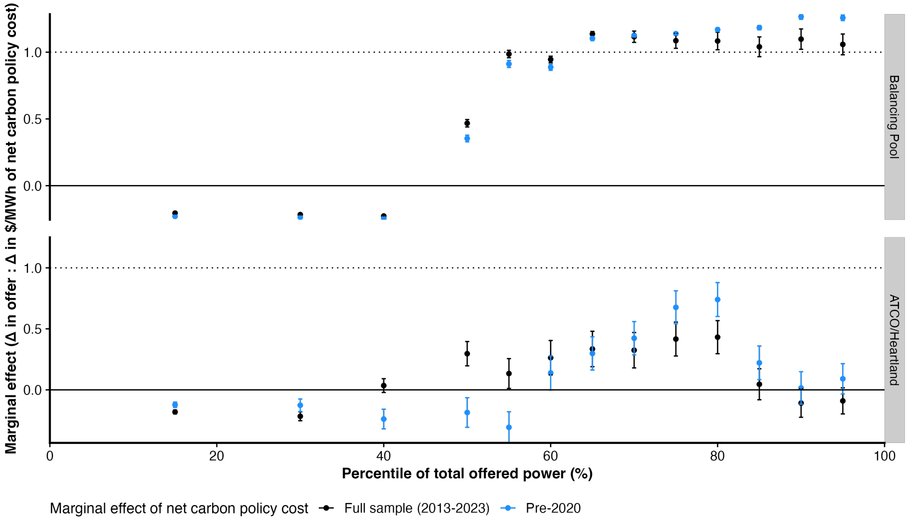


The Impact of the Balancing Pool Period

Facility	Plant Owner	Unit ID	Capacity (MW)	End of PPA Term	Returned to Balancing Pool	Conversion to gas-fired steam or retirement
Battle River	ATCO	BR3	147	Dec 31, 2013	NA	NA
		BR4	147	Dec 31, 2013	NA	Mar 8, 2022
		BR5	368	Dec 31, 2020	Jan 1, 2016	Nov 19, 2021
Genesee	Capital Power	GN1	381	Dec 31, 2020	NA	May 1, 2024*
		GN2	381	Dec 31, 2020	NA	July 1, 2024*
		GN3	466	NA	NA	July 12, 2024*
HR Milner	ATCO	HRM	144	Jan 01, 2013	NA	May 8, 2020
Keephills	TransAlta	KH1	383	Dec 31, 2020	May 5, 2016	Jan 11, 2022
		KH2	383	Dec 31, 2020	May 5, 2016	July 21, 2021
		KH3	466	NA	NA	Jan 11, 2022*
Sheerness	ATCO (50%)	SH1	378	Dec 31, 2020	Mar 7, 2016	July 30, 2021
	TransAlta (50%)	SH2	378	Dec 31, 2020	Mar 7, 2016	July 30, 2021
Sundance A	TransAlta	SD1	280	Dec 31, 2020	Mar 7, 2016	Jan 1, 2018*
		SD2	280	Dec 31, 2020	Mar 7, 2016	July 31, 2018*
Sundance B	TransAlta	SD3	353	Dec 31, 2020	Mar 7, 2016	July 31, 2020*
		SD4	353	Dec 31, 2020	Mar 7, 2016	Apr 1, 2022*
Sundance C	TransAlta	SD5	353	Dec 31, 2020	Mar 24, 2016	Nov 1, 2021*
		SD6	357	Dec 31, 2020	Mar 7, 2016	Feb 1, 2021

Note: * denotes a retirement as opposed to a conversion. Supercritical coal units at Keephills (KH3, 466MW, converted to gas Jan 11, 2022) and Genesee (GN3, 466 MW, converted to gas July 12, 2024) were never part of the PPA system, as they were built as merchant generators. A more extensive re-power of the Genesee 1 and 2 units saw them reclassified as natural gas combined-cycle plants after the retirements listed above.

Hints from the Balancing Pool's offers



Discussion

- Alberta's energy-only power market provides an ideal environment to study carbon price pass-through
- Results provide no support for full-cost pass-through in any hours
- Results suggest that strategic considerations are overwhelming carbon price pass-through incentives
- What's missing from the empirical work: ML estimates with right and left censoring of true opportunity costs of power
- Confounding factors I'm not sure we can deal with despite the wealth of data
- Questions for additional research

Contact info

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